

Lab 3: Neuro - Measurement Station 2

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October 16, 2025

Task 1: Series Resistor Measurement

$R_s = 2.17$ kOhm (multimeter measurement), R_{smd} nominal: 4.701 kOhm. Assembled circuit with R_s in series with R_{smd} . R_{smd} measured: 4.663 kOhm. Value close to nominal, circuit validated. Code and circuit functioning.

Visuals

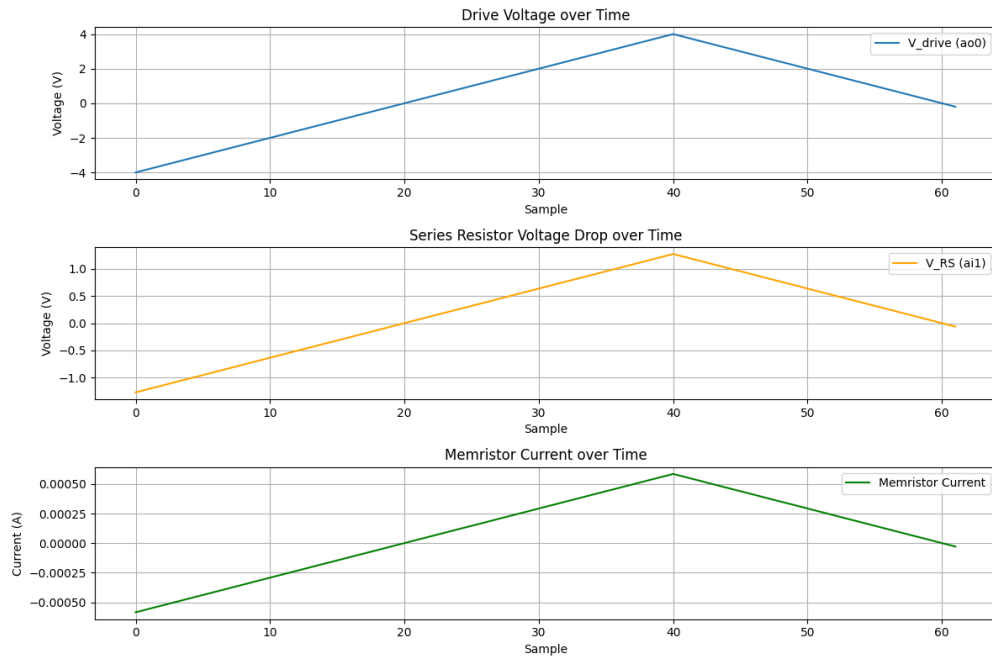


Figure 1: Task 1: Time Series Plot

Task 2: I(V) Curve Measurement

Observed expected I(V) curve with characteristic jumps.

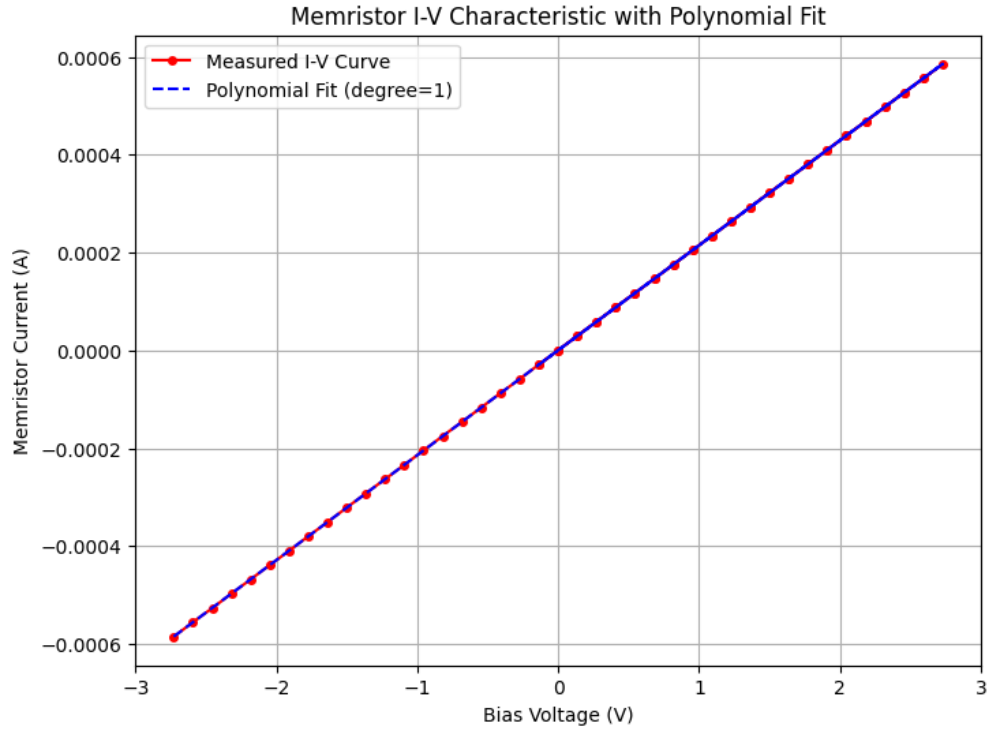


Figure 2: Task 1: Fit Plot

Visuals

Task 3: $R(T)$ Characteristics Measurement Setup

User interface, VISA/SCPI control, and data acquisition setup completed for $R(T)$ measurements. Calculations for sample resistance and temperature implemented. Data logging and plotting updated. Power supply ON/OFF controls added. Remember to update the 'PeltierPowerSupplyVisaResource'.

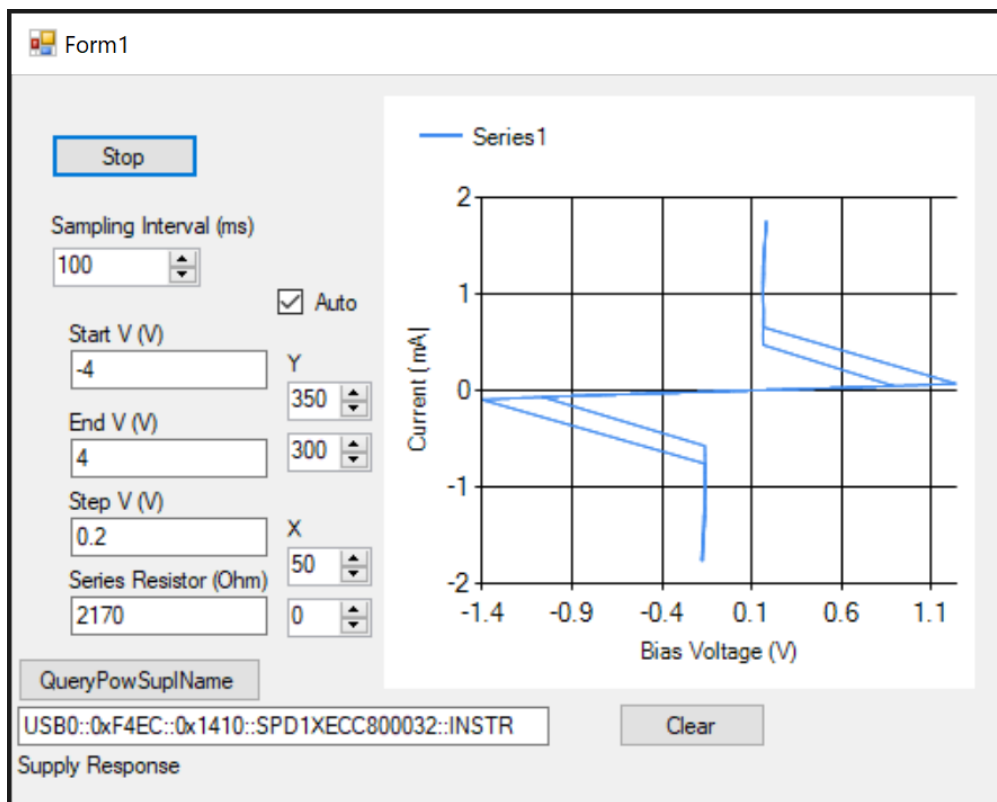


Figure 3: Task 2: I(V) Curve Screenshot